



Indywidualny Plan Badawczy (IPB) Individual Research Plan (IRP)

1. Podstawowe dane / Basic data

Podstawowe informacje dotyczące doktoranta i realizowanej rozprawy doktorskiej. W przypadku promotora spoza Politechniki Wrocławskiej należy podać afiliację. / *Basic information on the doctoral student and the doctoral dissertation underway. In the case of a supervisor from outside Wrocław University of Science and Technology, affiliation should be provided.*

Imię i nazwisko / Name, surname	Grzegorz Piotrowski
Dyscyplina kształcenia / Discipline of education	informatyka techniczna i telekomunikacja / information and communication technology
Wydział / Faculty	W4 Wydział Informatyki i Telekomunikacji / Faculty of Information and Communication Technology
Katedra / Department	K46 Katedra Sztucznej Inteligencji / Department of Artificial Intelligence
Data rozpoczęcia kształcenia / Date of commencement of the doctoral education	01.10.2025
Nr ORCID	0009-0004-6013-0461
Promotor/ promotorzy / Supervisor(s)	dr hab. inż. Tomasz Jan Kajdanowicz, prof. uczelni, Politechnika Wrocławska (K46, W4), ORCID 0000-0002-8417-1012
Promotor pomocniczy / Assistant supervisor	—

2. Temat rozprawy doktorskiej / Topic of doctoral dissertation

Propozycja tematu rozprawy doktorskiej. Przedstawiony temat nie musi stanowić ostatecznego tytułu, jaki zostanie nadany rozprawie doktorskiej złożonej na zakończenie kształcenia w Szkole Doktorskiej. / *Proposed topic for a doctoral dissertation. The presented topic does not have to constitute the final title of the doctoral dissertation submitted at the end of education at the Doctoral School.*

PL: Efektywne pod względem danych przenoszenie polityk nawigacji robotów mobilnych z symulacji do rzeczywistości z wykorzystaniem neuronowej rekonstrukcji scen (podejście real-to-sim-to-real)

EN: Data-efficient sim-to-real transfer of mobile-robot navigation policies using real-to-sim-to-real neural scene reconstruction

3. Harmonogram przygotowania rozprawy doktorskiej / Schedule for the preparation of the doctoral dissertation

Harmonogram przygotowania rozprawy doktorskiej z wyszczególnieniem etapów i miejsc prowadzonych badań, w tym terminów zakończenia badań cząstkowych (semestr kształcenia) i opracowania ich wyników oraz opcjonalnie w zależności od rodzaju prowadzonych badań, wybranych między innymi spośród następujących:

- realizacja badań (w tym przygotowanie stanowiska badawczego, przygotowanie materiałów do badań, wykonanie pomiarów/eksperymentów, opracowanie otrzymanych wyników),



- przygotowanie i wysłanie artykułu naukowego do redakcji czasopisma naukowego,
- wysłanie zgłoszenia patentowego do właściwego urzędu,
- opracowanie propozycji wdrożenia dla przemysłu,
- udział w konferencjach, warsztatach lub szkołach letnich/zimowych i prezentacja wyników badań,
- przygotowanie wniosku i wystąpienie o grant,
- mobilność krajowa lub zagraniczna doktoranta (w tym konsultacje naukowe i wizyty studyjne w ośrodkach krajowych lub zagranicznych).

A schedule for the preparation of the doctoral dissertation detailing the stages and locations of the research to be conducted, including the dates for completing the sub-studies and developing their results, and optionally, depending on the type of research to be conducted, selected among the following:

- *execution of the research (including preparation of the research site, preparation of materials for the research, execution of measurements/experiments, elaboration of the obtained results),*
- *preparation and submission of a scientific article to the editor of a scientific journal,*
- *submission of a patent application to the relevant office,*
- *development of an implementation proposal for the industry,*
- *participation in conferences, workshops or summer/winter schools and presentation of research results,*
- *preparation of an application and application for a scholarship,*
- *domestic or foreign mobility of the doctoral student (including scientific consultations and study visits to domestic or foreign centres).*

Okres realizacji zadania (semestr kształcenia) / Task implementation period (semester)	Zwięzły opis zadania / Brief description of the task
Semestr 1	<i>(Oct 2025 – Feb 2026)</i> Preparatory phase. T1.1 Literature review: sim-to-real transfer, domain randomization, neural scene reconstruction (NeRF, 3D Gaussian Splatting), learning-based navigation. <i>D</i> : annotated bibliography and gap statement (basis of §6). Place: K46.
Semestr 2	<i>(Mar – Sep 2026)</i> T2.1 Research questions, hypotheses and methods; survey of reconstruction and simulation tools (candidates in §9). <i>D</i> : this IPB, agreed with the supervisor and signed by 30.09.2026. Place: K46.



Semestr 3	(Oct 2026 – Feb 2027) Stage I – real-to-sim pipeline (RQ1, H1). T3.1 Tool selection (simulator, reconstruction method) and capture-to-simulator pipeline (SfM + 3DGS + collision mesh); fidelity validation on ≥ 2 indoor scenes; first policies trained in reconstructed vs. generic scenes (simulation only). <i>D:</i> tool-selection note, pipeline code, fidelity report (PSNR/SSIM/LPIPS, geometric error). Place: K46, WCSS. T3.2a Test stand: written agreement on robot access (planned cooperation with the K29 Denali laboratory, or another PWR platform) and sensor decision (SzD Minigrant application if a sensor must be bought). <i>D:</i> agreement, by Nov 2026. T3.2b Pilot deployment of a policy trained in one reconstructed scene; ROS 2 logging. <i>D:</i> pilot results. Place: Denali. <i>Decision point:</i> without robot access by end of Jan 2027, the pilot moves to the start of sem. 4 and Stage I is also evaluated offline on public real-world navigation datasets. T3.3 Prepare and submit an article to IEEE RA-L (fallback: IROS 2027, deadline 1 Mar 2027): Stage I pipeline, fidelity validation and simulation results; the real-robot pilot is included if available but is not a precondition. <i>D:</i> submission, February 2027 (§11).
Semestr 4	(Mar – Sep 2027) Stage II – policy learning and real-robot transfer (RQ1–RQ2, H1, H4). T4.1 Train RL / imitation-learning policies in reconstructed scenes; real-robot evaluation of H1 against the generic-simulator and real-data-only baselines in ≥ 2 scenes, plus the first real-data-budget curve (RQ2). <i>D:</i> experimental dataset and analysis of results (H1 completed). Place: K46, Denali, WCSS. T4.2 Grants and international activity: NCN PRELUDIUM application (call expected spring 2027); poster at an international summer/winter school; invitation of a foreign co-author for the Stage II article; host-lab choice and NAWA Bekker application for the semester-6 visit. <i>D:</i> submitted applications, poster. T4.3 Preparation of the mid-term evaluation. <i>D:</i> autoreferat (≤ 5 pages) and 15-min presentation (deadline ~mid-Oct 2027).
Semestr 5	(Oct 2027 – Feb 2028) Mid-term evaluation (Nov 2027). Stage III – closing the loop (RQ4, H3, H4). T5.1 Paired sim/real rollouts; SRCC of reconstructed vs. generic simulation. <i>D:</i> validation report (H3 completed). T5.2 Real-to-sim correction of appearance and dynamics from a few real rollouts; data-budget comparison with real-data-only training (H4). <i>D:</i> analysis of results. Place: K46, Denali. T5.3 Article with Stage II results submitted to RSS 2028 or IEEE RA-L. <i>D:</i> submission.



Semestr 6	(Mar – Sep 2028) Stage IV – generalization (RQ3, H2). T6.1 Augmentation of reconstructed scenes (object rearrangement, lighting, sensor noise, dynamic obstacles); evaluation in real environments that were not captured. <i>D:</i> ablation study (H2 completed). T6.2 Foreign research visit (3 months, NAWA Bekker; if not funded, a 5–30-day Erasmus+ short-term mobility to a European lab; hosts in §12): experiments in new environments / on a second robot. <i>D:</i> visit report, joint experiments. T6.3 Article with Stage III results submitted to IROS 2028 or IEEE RA-L. <i>D:</i> submission.
Semestr 7	(Oct 2028 – Feb 2029) Stage IV – consolidation. T7.1 Full comparison study (real-data budget vs. success rate, all baselines). <i>D:</i> final results (H4 completed). T7.2 Release of code, reconstructed scenes and real-world evaluation data. <i>D:</i> public repository. T7.3 Journal article summarizing the method (Robotics and Autonomous Systems or IEEE RA-L); start of dissertation writing. <i>D:</i> submission, dissertation outline.
Semestr 8	(Mar – Sep 2029) T8.1 Editing of the doctoral dissertation; final version. <i>D:</i> dissertation submitted by 30.09.2029 (§4). T8.2 Responses to reviews of pending articles. <i>D:</i> revised manuscripts.

4. Termin złożenia rozprawy doktorskiej / Date of the doctoral dissertation submission

Planowany termin złożenia rozprawy doktorskiej nie może przekraczać daty zakończenia ósmego semestru kształcenia w Szkole Doktorskiej. / A planned date for submission of the dissertation, which must not exceed the date of completion of the eighth semester of education in the Doctoral School.

30.09.2029

(rekomendowana data to ostatni dzień z okresu 4 lata kształcenia w Szkole Doktorskiej, np. jeżeli kształcenie rozpoczęło się 1.10.2025, to proszę wpisać 30.09.2029 /

The recommended date, which marks the final day of the four-year doctoral training period. For example, if your training began on October 1st, 2025, please enter September 30th, 2029)

5. Uzasadnienie wyboru tematu rozprawy doktorskiej / Justification for choosing the topic of the doctoral dissertation

Opis zawierający uzasadnienie wyboru tematu rozprawy doktorskiej uwzględniający potencjalne obszary zastosowania planowanych wyników rozprawy doktorskiej (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description containing the justification for the selection of the topic of the doctoral dissertation, taking into account the potential areas of application of the planned results of the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Autonomous mobile robots such as service robots, warehouse and inspection platforms, and delivery vehicles depend on perception-driven navigation policies. Modern learning-based approaches (deep reinforcement learning, imitation learning, large pretrained navigation models) outperform hand-engineered pipelines in unstructured scenes, but they are extremely data-hungry. Near-perfect point-



goal navigation in simulation took billions of frames of experience. Collecting data at that scale on real robots is slow, expensive, and risky for the hardware and for people.

Simulation is the natural alternative, but policies trained in simulation often fail when deployed on a real robot. This is the **sim-to-real gap**: differences in appearance (textures, lighting, sensor noise), geometry, and dynamics (friction, actuation delays). The standard remedy, domain randomization, requires careful manual design and still gives no guarantee for a particular target environment. The opposite option, training directly on real data, does not scale.

Recent progress in **neural scene reconstruction** (Neural Radiance Fields, 3D Gaussian Splatting) makes a third path possible: **real-to-sim-to-real**. A few minutes of real-world capture are turned into a photorealistic, interactive simulation of the target environment. The policy is trained there at scale and then deployed back on the real robot. This shifts the cost from collecting *robot experience* to collecting *cheap scene captures*. It is a promising way out of the data bottleneck, but it has not yet been studied systematically for mobile-robot navigation. Open questions include how much real data is really needed, how to generalize beyond the captured scenes, and how to correct the simulation with feedback from real rollouts.

Why this topic, and why in this discipline. The core problems are problems of *information and communication technology*: learning algorithms, data-efficient training methodology, software for building simulators from sensor data, and methods for evaluating whether simulation results can be trusted. The robot is the test platform, not the object of the research. The topic fits the machine learning, computer vision and representation learning work of the Department of Artificial Intelligence (K46), and the computation-heavy parts (3D reconstruction, large-scale policy training) can run on the GPU infrastructure available to PWr researchers (WCSS, PLGrid). Its main result, a measured answer to "how much real data is enough", is useful to both researchers and practitioners, whatever the outcome.

Potential application areas: indoor service and assistive robotics, logistics and warehouse automation, inspection of industrial facilities, last-mile delivery, and rapid deployment of robots in new buildings with minimal on-site data collection. The methods also carry over to other embodied systems, such as manipulators and drones, and to digital twins of real environments more generally.

6. Zarys aktualnego stanu badań w tematyce rozprawy doktorskiej / **Outline of the current state of research in the topic of the doctoral dissertation**

Przegląd literaturowy zawierający opis aktualnego stanu badań w tematyce rozprawy doktorskiej uwzględniający odwołania do najważniejszych pozycji literaturowych związanych z tematyką rozprawy doktorskiej (maks. 2 strony, czcionka 11 pt., interlinia 1). / A literature review containing a description of the current state of research in the subject of the doctoral dissertation, including references to the most important literature items related to the subject of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

Learning-based navigation and its data demands. Photorealistic simulators such as Habitat [1, 2] and CARLA [3] made large-scale training of embodied agents possible. Wijmans et al. [4] reached near-perfect point-goal navigation, but only after 2.5 billion frames of simulated experience, a volume out of reach on physical robots. *Navigation foundation models* (GNM [5], ViNT [6], NoMaD [7]) are trained on pooled real-robot datasets, and Open X-Embodiment [8] shows that robot learning scales with data. Collecting such data is still costly, and every new environment or platform typically needs more real-world data.

The sim-to-real gap. Policies trained in simulation degrade on real hardware because of differences in appearance, geometry and dynamics [9, 10]. Kadian et al. [11] showed that simulation performance can be a poor predictor of real-world performance unless the simulator is carefully tuned, and introduced the Sim-vs-Real Correlation Coefficient (SRCC) to measure it. Truong et al. [12] found that



lower-fidelity simulation with abstracted dynamics can transfer better for navigation. So *which* aspects of reality to model is itself an open question.

Domain randomization and adaptation. Domain randomization trains policies on widely varied synthetic appearance [13, 14] or dynamics [15], so that reality looks like just another variation. It powers landmark results in agile flight [16] and drone racing [17]. Its drawbacks are hand-tuned randomization ranges and robustness bought at the cost of performance in the specific target domain. Adaptive approaches fit simulation parameters to real rollouts [18], and domain adaptation [19] reduces the amount of real data needed, but does not remove the need for it.

Neural scene reconstruction and real-to-sim-to-real. Neural Radiance Fields [20] and 3D Gaussian Splatting (3DGS) [21] produce photorealistic novel views from ordinary image captures, and 3DGS renders in real time. This allows *digital-twin* simulators built from real data: NeRF2Real [22] trained bipedal skills inside a NeRF of the target scene, and RialTo [23] robustified manipulation policies in digital twins built from small amounts of real data. Since 2024 the idea has reached navigation: 3DGS simulators gave direct sim-to-real transfer of visual drone navigation [24–26]; VR-Robo [27] transferred RGB-only legged goal-reaching policies from a 3DGS twin with mesh-based physics; Vid2Sim [28] built urban simulators from monocular video and reported a 68.3% real-world success-rate gain over prior simulators; ReaDy-Go [29] added animated humans as moving obstacles. Closest to this dissertation, EmbodiedSplat [30] fine-tunes pretrained image-goal navigation policies in meshes reconstructed from 20–30-minute phone captures of indoor scenes, improves real-robot success rate in a captured scene, and reports high SRCC for the reconstructed meshes. These navigation systems use a fixed capture per scene: none varies the amount of real data (capture length, real rollouts) or compares against policies trained on real robot experience at matched budgets. The closest such analysis, in manipulation, is RialTo's ablation over 0–15 real demonstrations [23]. Multi-room scenes, collision geometry and generalization to environments that were never captured also remain open.

Research gap. There is no systematic study of the **trade-off between real-data budget and deployed navigation performance** in a real-to-sim-to-real pipeline. We also lack methods that (i) combine reconstructed scenes with targeted randomization to generalize beyond the captured environments, and (ii) use a small number of real rollouts to iteratively correct the simulation. This dissertation addresses that gap.

References

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- [3] A. Dosovitskiy et al., "CARLA," CoRL, 2017.
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- [19] K. Bousmalis et al., "Using Simulation and Domain Adaptation to Improve Efficiency of Deep Robotic Grasping," *ICRA*, 2018.
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- [24] A. Quach et al., "Gaussian Splatting to Real World Flight Navigation Transfer with Liquid Networks," *arXiv:2406.15149*, 2024.
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- [28] Z. Xie et al., "Vid2Sim," *CVPR*, 2025.
- [29] S. Yoo et al., "ReaDy-Go," *IEEE RA-L*, 2026.
- [30] G. Chhablani et al., "EmbodiedSplat," *ICCV*, 2025.

7. Pytania i hipotezy badawcze / Research questions and hypotheses

Opis najważniejszych pytań i hipotez badawczych związanych z realizowaną rozprawą doktorską (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description of the most important research questions and hypotheses related to the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Thesis. A photorealistic, interactive simulation reconstructed from a few minutes of real-world capture of the target environment lets learned mobile-robot navigation policies reach real-world performance not worse than policies trained on real robot experience, while needing at least ten times less real-world data.

Core and extensions. The dissertation stands on **RQ1–RQ2 with H1 and H4** (the real-data-budget study). RQ3/H2 (generalization) and RQ4/H3 (closed-loop correction, predictivity) are extensions.

RQ1. Can a simulation reconstructed from minutes of RGB(-D) video of the target environment (*real-to-sim*), with policies deployed back on the robot (*sim-to-real*), close the visual and geometric gap for learned navigation?

RQ2. How does deployed performance scale with the real-data budget, and what budget suffices compared with training on real robot experience?

RQ3. How should reconstructed scenes be augmented so that policies generalize to environments that were *not* captured?

RQ4. Can a few real rollouts iteratively correct the simulation (appearance, dynamics)?

Common protocol. 20 fixed start–goal pairs $\times$ 3 trials per policy and environment (60 episodes). Metrics: success rate (SR; primary), SPL, collisions. *Real-data budget* B = minutes of real data collection (capture plus robot rollouts). The real-data-only baseline is a pretrained navigation model fine-tuned by behaviour cloning on teleoperated trajectories of budget B; the generic-simulator baseline uses a public scene dataset of the simulator chosen in T3.1. Tests are one-sided ($\alpha = 0.05$) with bootstrap confidence bounds over start–goal pairs and Holm correction within each hypothesis. All values are pre-registered in the project repository; changes are logged with reasons.

H1 (RQ1, core). A policy trained only in the reconstruction of the target environment (a) reaches a higher real-world SR than the same policy trained in a generic simulator with domain randomization, and (b) is **non-inferior** to a policy trained on real robot data with a $\geq 10\times$ larger budget: the lower 95% confidence bound of $SR_{recon} - SR_{real}$, pooled over ≥ 2 environments (120 episodes per policy), is above -15 pp. This gives $\approx 90\%$ power at $SR \approx 0.8$, rechecked after the pilot.



H2 (RQ3). In uncaptured environments, reconstruction + targeted augmentation reaches a higher SR than both reconstruction alone and generic domain randomization alone.

H3 (RQ4). Over ≥ 10 policy variants, the Sim-vs-Real Correlation Coefficient (SRCC) of the reconstructed simulation is higher than that of a generic simulator (95% CI of the difference excludes 0) and does not drop after real-rollout refinement (lower 95% CI bound of the change ≥ -0.1).

H4 (RQ2, core). The full pipeline, including refinement from a few real rollouts, reaches a target SR τ (best real-data-only SR minus 10 pp) with $B_{\text{pipeline}}(\tau) \leq 0.1 \cdot B_{\text{real}}(\tau)$: the upper 95% CI bound of the ratio is ≤ 0.1 . If real-data-only training misses τ within the largest feasible budget B_{max} , the ratio is reported as a bound. Otherwise the budget curve still answers RQ2.

8. Wkład spodziewanych wyników dla rozwoju dyscypliny naukowej / Contribution of the expected results to development of scientific discipline

Krótki opis spodziewanego wkładu wyników rozprawy doktorskiej dla rozwoju dyscypliny naukowej, w której realizowana jest rozprawa doktorska (maks. 1 strona, czcionka 11 pt., interlinia 1). / Brief description of the expected contribution of the results of the doctoral dissertation to the development of the scientific discipline in which the doctoral dissertation is carried out (at most 1 page, font 11, line spacing 1).

The dissertation is carried out in **information and communication technology** (*informatyka techniczna i telekomunikacja*). Its contributions are learning algorithms, data-efficient training methodology, software, data and evaluation methodology for machine learning in autonomous systems. The robot is the test platform; control design and mechanical engineering are not the object of the research.

Key original contribution (core, RQ1–RQ2, H1 + H4). The first systematic, quantitative characterization, to our knowledge, of how the deployed performance of learned mobile-robot navigation policies depends on the **real-data budget** in a real-to-sim-to-real pipeline. It answers "how much real data is enough" with measured budget–success-rate curves and a non-inferiority comparison against policies trained on real robot experience. This is the result the dissertation stands on.

Contributions, in ITiT terms:

1. **Data-efficient training methodology (core; RQ2, H4).** A procedure for training navigation policies under an explicit real-data budget. It covers scene capture and a small number of real rollouts, and the budget–performance curves that quantify the trade-off.
2. **Software pipeline (core; RQ1, H1).** A reproducible, open real-to-sim-to-real pipeline that turns short RGB(-D) captures into interactive, photorealistic training environments (3D Gaussian Splatting for appearance, extracted geometry for collisions) inside a GPU simulator.
3. **Learning algorithm for generalization (extension; RQ3, H2).** A scheme for augmenting reconstructed scenes so that policies transfer to environments that were never captured. It connects the digital-twin and domain-randomization lines of research.
4. **Closed-loop simulation correction and evaluation methodology (extension; RQ4, H3).** A method to update simulation parameters from a few real rollouts, and a protocol for measuring how well simulation predicts real performance (SRCC, pre-specified margins and decision rules), so that sim-to-real claims can be tested rather than asserted.
5. **Data and benchmark (supports all RQs).** Released code, reconstructed environments and a real-world evaluation set with fixed start–goal pairs, supporting reproducible sim-to-real benchmarking.

Significance for the discipline. Collecting robot experience is the main cost of learning-based robotics. Replacing most of it with cheap scene captures and simulation lowers the cost and time of



deploying learned systems in new environments. The evaluation methodology (item 4) is reusable for other simulators and embodied-AI tasks.

Dissemination. Results are planned for venues assigned to ITiT on the ministerial list (5.01.2024): IEEE RA-L (200 pts), RSS (200 pts), IEEE/RSJ IROS (140 pts) and Robotics and Autonomous Systems (140 pts), as in §3, §9, §11 and §12.

9. Planowane metody badawcze / Planned research methods

Krótki opis metod badawczych, które są będą użyte w czasie realizacji rozprawy doktorskiej dla (maks. 2 strony, czcionka 11 pt., interlinia 1). / Brief description of the research methods that will be used during the implementation of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

The research follows four stages (I–IV) matching §3; for each we give the method, data, metric and success criterion. Stage II (H1 and the real-data budget, RQ1–RQ2) is the core of the dissertation; Stages III–IV extend it.

Test stand and data. Real-robot experiments use a mobile-robot platform available at PWr (planned cooperation with the K29 "Denali" Autonomous Robots Laboratory, which operates Pioneer 3-DX and DrRobot Jaguar 4x4 robots under ROS 2; terms agreed in sem. 3, T3.2a). The robot carries an RGB-D camera and uses wheel odometry; if no suitable sensor is available, one is bought from an SzD Minigrant. A fixed capture protocol records short RGB-D video of each environment (minutes per scene) and a small number of teleoperated trajectories. The **real-data budget** (capture minutes, number of real rollouts) is an explicit experimental variable. *Estimated robot time* (at ~2 min per episode including reset): about 35 robot-hours in Stage II (6 policies × 120 episodes, plus teleoperated data for the real-data-only baseline) and about 20 in Stage III (≥ 10 variants × 60 episodes); resets are scripted and the H1 core comparisons are run first. **Fallback** if robot access is delayed: offline evaluation on public real-world navigation datasets (e.g. SCAND, RECON) with real-robot deployment moved to the next semester. Compute: WCSS Lem (NVIDIA H100) and PLGrid allocations (to be applied for), plus departmental GPUs.

Stage I – real-to-sim reconstruction (RQ1; sem. 3).

- *Method:* camera poses from Structure-from-Motion (COLMAP) or RGB-D SLAM; appearance from 3D Gaussian Splatting (NeRF as baseline); collision geometry from TSDF depth fusion or mesh extraction; import into a GPU simulator (Isaac Sim / Isaac Lab or Habitat, chosen in T3.1) with rendering from the reconstruction and physics from the mesh.
- *Data/equipment:* RGB-D captures of ≥ 2 indoor scenes at PWr (corridors, labs, offices).
- *Metric and success criterion:* PSNR, SSIM and LPIPS on held-out real views, and depth error against sensor depth. A scene passes if its held-out PSNR is within a margin X dB of the value reported for 3DGS [21] on comparable indoor scenes (X fixed in the repository before Stage II) and its collision mesh has no holes on the evaluation paths.

Stage II – policy learning and real-robot transfer (RQ1–RQ2, H1, H4; sem. 3–4).

- *Method:* (i) deep reinforcement learning (PPO / DD-PPO) for point-goal and image-goal navigation, trained at scale in the reconstructed scenes; (ii) imitation learning and fine-tuning of pretrained navigation models (GNM, ViNT, NoMaD) on trajectories generated in simulation. Input: RGB(-D); output: velocity commands.
- *Baselines:* the same policy trained in a generic simulator (public scene dataset, with and without domain randomization); a pretrained model fine-tuned by behaviour cloning on teleoperated real trajectories at matched and ≥ 10× larger budgets; pretrained models zero-shot.
- *Metric and success criterion:* success rate (primary), SPL, collisions and interventions under the §7 protocol, and success rate as a function of the real-data budget B; H1 decision rules in §7. The real-robot pilot (one scene) runs in sem. 3 if robot access is agreed, otherwise at the start of sem. 4; the H1 study in ≥ 2 environments is completed by the end of sem. 4 (before the mid-term).



Stage III – closing the loop and sim-to-real validation (RQ4, H3, H4; sem. 5).

- *Validation protocol:* paired simulated and real rollouts on the same start–goal pairs. Over ≥ 10 policy variants we compute the **Sim-vs-Real Correlation Coefficient (SRCC)** between simulated and real success/SPL for (a) the reconstructed and (b) the generic simulator, and relate per-scene fidelity (Stage I metrics) to the transfer gap.
- *Method:* the discrepancy between real and replayed trajectories is used to update the scene (re-capture of poorly reconstructed regions) and to fit dynamics and sensor parameters (system identification, adaptive randomization in the style of SimOpt); the policy is then fine-tuned in the corrected simulation. The number of real rollouts per iteration is small and reported.
- *Metric and success criterion:* SRCC with bootstrap confidence intervals; real rollouts needed to reach the target success rate compared with real-data-only training; H3 and H4 decision rules in §7.
- *Fallback* (low SRCC or no gain): reported as a negative finding, with the error attributed to appearance, geometry or dynamics by ablation; Stage II conclusions do not depend on this stage.

Stage IV – generalization and consolidation (RQ3, H2, H4; sem. 6–7).

- *Method:* augmentation of reconstructed scenes: object insertion and rearrangement, lighting and appearance randomization, sensor noise and latency, dynamics randomization, simulated pedestrians. Ablation: generic simulator + domain randomization vs. reconstruction only vs. reconstruction + augmentation.
- *Data/equipment:* captured (target) and **uncaptured** (unseen) real environments at PWR and, during the foreign visit, at the host lab.
- *Metric and success criterion:* success rate and SPL in unseen environments (H2 decision rule in §7); the full real-data-budget curve over all baselines.

Statistics, reproducibility and dissemination. All real-robot comparisons follow the §7 protocol. Code, configurations and scenes are versioned, logged and released where permitted. Each stage is published in listed venues: IEEE RA-L, RSS, IROS or Robotics and Autonomous Systems (§3, §11).

10. Streszczenie popularnonaukowe / Abstract for general public

Streszczenie popularnonaukowe należy przygotować w dwóch językach: polskim i angielskim. Wersje językowe muszą być tożsame. Opis powinien być napisany językiem popularnonaukowym i zawierać cel rozprawy, opis planowanych badań naukowych oraz powodów, dla których podjęta została ta tematyka badawcza, a także najważniejsze spodziewane efekty (maks. 1 strona, czcionka 11 pt., interlinia 1). / The description for general public must be in Polish and in English. The language versions must be identical. The description should be written for general public and should include the doctoral dissertation goal, description of research, reasons for attempting a particular research topic and substantial results expected (at most 1 page, font 11, line spacing 1).

Streszczenie popularnonaukowe

Roboty mobilne, na przykład w magazynach, szpitalach czy biurach, coraz częściej uczą się poruszać samodzielnie dzięki metodom sztucznej inteligencji. Takie uczenie wymaga jednak ogromnej liczby przykładów, często milionów prób i błędów. Zebranie ich na prawdziwym robocie trwa bardzo długo, jest kosztowne i może prowadzić do uszkodzeń sprzętu. Dlatego roboty uczą się zwykle w symulacji komputerowej, czyli w wirtualnym świecie przypominającym grę. Problem polega na tym, że świat wirtualny różni się od prawdziwego: inaczej wyglądają ściany, światło i przedmioty, a robot zachowuje się inaczej na śliskiej podłodze. Robot wyszkolony w symulacji często gubi się po przeniesieniu do rzeczywistości.

Celem rozprawy jest opracowanie metody, która pozwoli nauczyć robota mobilnego nawigacji w konkretnym miejscu przy użyciu bardzo małej ilości prawdziwych danych. Pomysł polega na nagraniu krótkiego filmu z danego miejsca, na przykład korytarza lub biura. Na jego podstawie metody sztucznej inteligencji odtwarzające trójwymiarowe sceny tworzą wierną, fotorealistyczną cyfrową



kopię tego miejsca. W tej kopii robot może bezpiecznie i szybko ćwiczyć miliony razy, a następnie wrócić do prawdziwego świata. Informacje z kilku prawdziwych przejazdów posłużą dodatkowo do poprawiania symulacji tak, by coraz lepiej odpowiadała rzeczywistości. Metoda zostanie sprawdzona na prawdziwym robocie w budynkach Politechniki Wrocławskiej i porównana z robotami uczonymi w zwykłej symulacji.

Badania pokażą, ile prawdziwych danych naprawdę potrzeba, aby robot działał niezawodnie, oraz jak sprawić, by radził sobie również w miejscach, których wcześniej nie widział. Spodziewanym efektem jest znaczne obniżenie kosztu i czasu wdrażania autonomicznych robotów w nowych budynkach, co może przyspieszyć ich wykorzystanie w logistyce, opiece i przemyśle. Opracowane oprogramowanie i dane zostaną udostępnione publicznie.

Abstract for general public

Mobile robots, for example in warehouses, hospitals or offices, increasingly learn to move on their own using artificial intelligence. This learning needs an enormous number of examples, often millions of trials and errors. Collecting them on a real robot takes a very long time, is expensive and can damage the hardware. That is why robots are usually trained in a computer simulation, a virtual world similar to a video game. The problem is that the virtual world differs from the real one: walls, light and objects look different, and the robot behaves differently on a slippery floor. A robot trained in simulation often gets lost when moved to reality.

The goal of this dissertation is to develop a method that teaches a mobile robot to navigate a specific place using very little real data. The idea is to record a short video of a place, such as a corridor or an office. From this video, artificial intelligence methods that reconstruct three-dimensional scenes create a faithful, photorealistic digital copy of the place. In this copy the robot can practice safely and quickly millions of times, and then return to the real world. Information from a few real runs will also be used to correct the simulation so that it matches reality more and more closely. The method will be tested on a real robot in the buildings of Wrocław University of Science and Technology and compared with robots trained in an ordinary simulation.

The research will show how much real data is truly needed for a robot to work reliably, and how to make it cope with places it has never seen before. The expected result is a significant reduction in the cost and time of deploying autonomous robots in new buildings, which may speed up their use in logistics, care and industry. The software and data developed will be made publicly available.

11. Termin oddania do druku artykułu naukowego lub monografii / Date of submission for publication of at least one scientific article or monography

Termin wystania do recenzji artykułu naukowego lub monografii naukowej zgodnie z wymaganiami art. 186 ust. 1 pkt 3 Ustawy stawianymi osobie ubiegającej się stopień doktora. Termin dotyczy oddania do druku co najmniej jednego artykułu naukowego opublikowanego w czasopiśmie naukowym lub w recenzowanych materiałach z konferencji międzynarodowej w odpowiedniej dyscyplinie naukowej, które w roku opublikowania artykułu w ostatecznej formie były ujęte w wykazie („lista ministerialna”) sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. b) lub 1 monografii naukowej wydanej przez wydawnictwo, które w roku opublikowania monografii w ostatecznej formie było ujęte w wykazie sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. a). / A date of submission for publication of at least one scientific article published in a scientific journal or in the peer-reviewed materials of an international conference in the relevant scientific discipline, which in the year of publication of the article in its final form were included in the list drawn up in accordance with the regulations issued under the Act (Art. 267 sec. 2 point 2(b) or 1 scientific monograph published by a publishing house, which in the year of publication of the monograph in its final form was included in the list drawn up in accordance with the regulations issued pursuant to the Act (Art. 267 sec. 2 point 2(a), referred to in Art. 186 sec. 1 point 3 of the Act.



February 2027 / luty 2027

Article submitted for review to **IEEE Robotics and Automation Letters (RA-L)**, ISSN 2377-3766, with a planned presentation at IEEE/RSJ IROS 2027 (if the RA-L-IROS window is offered). It reports the Stage I pipeline and its validation (T3.3); the real-robot pilot is included if available but is not required. RA-L is worth **200 points** on the ministerial list and is assigned to the discipline *informatyka techniczna i telekomunikacja*.

12. Inne / Other comments

Dodatkowe wyjaśnienia, komentarze, podsumowanie związane z indywidualnym planem badawczym (maks. 1 strona, czcionka 11 pt., interlinia 1). / Additional explanations, comments, summary related to the individual research plan (at most 1 page, font 11, line spacing 1).

Planned outputs by year (points: ministerial list of 5.01.2024; re-checked against the 2027 list).

Year	Papers (venue, points)	Grants	Mobility, events
1 (2025/26)	—	—	—
2 (2026/27)	Stage I: IEEE RA-L (200), Feb 2027 (§11)	SzD Minigrant; NCN Preludium	Summer/winter school poster
3 (2027/28)	Stage II: RSS 2028 (200) or RA-L (200); Stage III: IROS 2028 (140) or RA-L (200)	NAWA Bekker	Mid-term; 3-month visit (sem. 6)
4 (2028/29)	Summary: Robotics and Autonomous Systems (140) or RA-L (200)	—	Code and data release; dissertation

Foreign research visit (candidate hosts, not yet contacted). (1) Multi-robot Systems Lab, Stanford University (M. Schwager): Gaussian-splatting scene models and navigation. (2) Z. Kira's group, Georgia Tech: EmbodiedSplat [30]. A European host, which also opens the Erasmus+ short-term route (5–30 days), is sought with the supervisor in sem. 4, together with a foreign co-author for the Stage II article.

Collaboration at PWr. K46 has no mobile-robot laboratory, so real-robot experiments are planned with the *Denali* Autonomous Robots Laboratory (K29, W12N; Pioneer 3-DX, Jaguar 4x4, ROS 2); a written agreement is task T3.2a (Nov 2026). Consultations on 3D reconstruction with K46's genwro.AI group are possible.

Risks and mitigation (semester affected).

- *No robot access or sensor* (3–4): decision point end of Jan 2027 (§3); public real-world datasets and simulation meanwhile; Minigrant/Preludium funds for a sensor; the ROS 2 pipeline can move to another robot.
- *Robot time exceeds lab access* (4–5): ~35 robot-hours in Stage II (§9); scripted resets, H1 core comparisons first, fewer policy variants.
- *Sim-to-real validation fails* (4–5): pre-specified SRCC protocol; a negative result still answers RQ2.
- *Poor reconstruction of low-texture or large scenes* (3–4): depth priors, scene splitting, mesh rendering.
- *Changing venue points* (3): re-check the 2027 list; keep a 140/200-point alternative.

Prior work (background only). A pre-PhD NLP paper (ACL 2025 Student Research Workshop) gives experience in representation learning; it is not part of the dissertation.

Ethics and data. Captures avoid or anonymize personal data (GDPR, PWr rules).



13. Określenie planowanej formy współpracy z promotorem / Planned form of cooperation with the supervisor

Krótki opis planowanej formy współpracy z promotorem, np. częstotliwość spotkań, forma spotkań i ich zakres, sposób komunikacji. / Brief description of the planned form of cooperation with the supervisor, e.g., frequency of meetings, form of meetings and their scope, method of communication.

- **Weekly meeting** (~1 h, in person at K46 or online via MS Teams) to discuss progress, results and next steps.
- **Monthly written progress summary** (completed tasks, results, problems, plan). It lists every IPB task from §3 with its % **completion**, in the same form as the task table of the mid-term self-report.
- **Semester review** against the IPB schedule (§3), before each annual report and the mid-term evaluation. Deviations from the plan and their reasons are recorded and agreed with the supervisor.
- **Supervisor co-signs** the annual reports and the mid-term self-report (autoreferat).
- **Paper-writing sprints** before each submission deadline, with joint editing in Overleaf. The supervisor co-authors and reviews the papers before submission.
- **Ongoing communication** via e-mail / Teams chat. The code and experiment logs are in a shared Git repository that the supervisor can access.

14. Opinia promotora pomocniczego / Opinion of assistant supervisor

Promotor pomocniczy (jeżeli jest wyznaczony) powinien zaopiniować IPB. / Assistant supervisor (if appointed) should provide an opinion of the IRP.

nie dotyczy / not applicable

.....
Podpis promotora pomocniczego /
Signature of assistant supervisor

15. Podpisy doktoranta i promotora/promotorów / Signatures of doctoral student and supervisor(s)

Wybierz datę/Select date

.....
Data / Date

.....
Podpis doktoranta / Signature of doctoral student

.....
Podpis promotora / Signature of supervisor

.....
Podpis drugiego promotora / Signature of second supervisor